

The Edge Grid: A Prototype for Decentralized, Verifiable Peer-to-Peer AI Inference

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Abstract—Centralized inference services concentrate pricing power, availability, and trust in a single operator, and decentralized physical infrastructure networks have already shown that peer-to-peer markets can sell connectivity, storage, and general compute at scale. Selling verifiable AI inference the same way raises a problem those networks do not solve: an anonymous edge provider, operating under a hard latency budget, has both the means and the incentive to return a plausible-looking wrong answer, and naive redundancy or centralized scheduling cannot detect that cheaply. We design a network in which a job is discovered and auctioned under a sealed-bid, warm-start-aware second-price rule; executed by a streaming edge inference runtime; committed to a Merkle-backed data-availability layer; and verified by a two-tier mechanism that separates a trustless cryptographic fraud proof from a sampled, quorum-voted semantic judge, with staked collateral slashed 80/20 between reporter and treasury on confirmed fraud. We build and measure a small-scale reference implementation on commodity CPU hardware: warm time-to-first-token averaged 610 ms with a 12× cold-start penalty; a sealed-bid auction cleared correctly across 57 trials and responded near-linearly to injected per-link network delay; content-addressed model weights were fetched, independently re-verified, and correctly rejected under three tampering scenarios; and on-chain settlement conserved value exactly across all three resolution paths. The verification layer's measured behavior is the paper's central finding: the deployable, edge-hostable judge configuration caught only 30–35% of two semantically subtle fraud strategies while scoring them confidently rather than hesitantly, and a follow-up experiment shows this failure tracks judge capability rather than the judge's relationship to the model it polices — refuting an explanation this evaluation initially favored.

Keywords—decentralized physical infrastructure networks, verifiable AI inference, LLM-as-a-Judge, second-price auctions, optimistic fraud proofs, staked settlement.

I. INTRODUCTION

Centralized inference services have made large-model access cheap and reliable at the cost of concentrating pricing power, infrastructure control, and data governance in a small number of operators. Decentralized physical infrastructure networks (DePIN) already sell connectivity, storage, and general compute competitively; none solve the narrower problem this paper addresses: **selling AI inference, rather than raw compute, from anonymous edge providers under a latency budget, with correctness**

verified cheaply enough that verification does not eat the economics it protects.

Redundant execution multiplies cost by the redundancy factor for every job, not just the audited fraction — economically incompatible with a centralized provider's marginal cost. Centralized verification reintroduces the single point of control the network exists to remove. Reputation punishes a *pattern*, not a specific transaction. What is needed is a mechanism that prices a latency budget at routing

time, verifies a *sampled* fraction of jobs cheaply, and makes fraud's economic consequence automatic — without a central adjudicator. Two properties make this harder than verifying storage: a wrong answer need not look wrong (a fabricated fact or negated claim can be fluent and indistinguishable from a correct one), and the party best positioned to catch a subtle semantic error is itself a language model — so the verification layer inherits every reliability question that attaches to language models generally, including that it can be confidently wrong. Section 5.4 is a direct measurement of that risk.

We design the network around a strict separation between two kinds of verification claim. A **cryptographic fraud proof** establishes with certainty that a provider served output inconsistent with what it committed on chain — a fact about two hashes, needing no challenge period. A **semantic verdict** is a judgment call about whether a faithfully-delivered output is actually correct, rendered by one or more independent validator models under a quorum rule. This lets the cheap, certain tier absorb one entire class of dishonesty for a fixed cost, and confines the expensive, probabilistic tier to the dishonesty that genuinely requires judgment.

Design contributions: (1) a market protocol combining a sealed-bid second-price auction with a

warm-start handicap applied to ranking score but never to payment, since model residency measures as a near-eligibility condition rather than a tradeable quality dimension under realistic latency budgets; (2) a two-tier verification protocol with a three-valued verdict that never collapses "unreachable" into either a pass or a fraud finding; (3) settlement as an escrow whose release requires no privileged party online, with unbonding collateral that stays slashable to close a front-running window.

Prototype and evaluation contributions: (1) a working, reproducible reference implementation with every figure traceable to a configuration-snapshotted run; (2) measurement that the deployable verification configuration is confidently wrong on two of four fraud strategies, and that this tracks judge *capability* rather than *lineage* — refuting an explanation we initially favored; (3) measurement that quorum aggregation reduces false positives well below individual members at no cost to recall; (4) identification of a methodological confound — a model-generated "honest" control is not reliably honest, and bounds every LLM-as-a-Judge precision figure — that recurred in our own pipeline even after being diagnosed once.

II. RELATED WORK

DePIN networks already sell compute, storage, or connectivity through token-incentivized provider sets: Bittensor rewards ML work but allocates reward predominantly by stake rather than output quality [1]; Akash [2] and Golem [3] auction general compute without an inference-specific verification layer; Gensyn [4] targets verifiable ML compute oriented toward training. None couples a latency-bounded auction to a sampled semantic verification layer with staked slashing.

Morpheus [5, 6] is the closest architectural precedent — an Arbitrum-L2 marketplace where providers bid and contracts match them to serve LLM inference. We differ in mechanism (a warm-start-aware second-price auction with hard-constraint eligibility) and in adding a two-tier verification layer outside Morpheus's stated scope. DGrid [7] and Parallax [8] describe decentralized inference serving without the auction or verification layers evaluated here; both are vendor litepaper/preprint, not peer-reviewed. **PolyLink** [9] is the closest refereed

academic parallel for settlement but does not evaluate an auction's response to injected latency (§5.3).

Our data-availability layer implements Celestia's *binding* property [10] against a local store, explicitly declining its *availability* guarantee. Settlement's challenge-window structure follows the fraud-proof lineage established for rollups [11, 10, 12]. The LLM-as-a-Judge paradigm [13, 14, 15] underlies our semantic tier; our contribution is a measured finding that judge *capability*, not *lineage* relative to the policed model, determines whether subtle falsehoods are caught (§5.4) — qualifying the intuition that same-family judges share blind spots. Petals [16, 17] addresses collaborative multi-peer inference, a complementary problem; Navigator [18] addresses decentralized *scheduling* alone, without verification or settlement.

No system surveyed combines Kademlia discovery, a warm-start-aware second-price auction under a hard latency budget, a two-tier verification pipeline separating a trustless fraud proof from a

capability-sensitive semantic check, and staked slashing, in one evaluated reference implementation.

III. SYSTEM DESIGN

System model. Four roles — requesters, providers, validators, and a settlement layer — compose a five-stage pipeline (discovery, market, inference, verification, settlement) communicating through a fixed message contract (Figure 1).

Threat model. Cheating providers are split into two trust classes: an output inconsistent with what was committed on chain is *provably* dishonest (a hash mismatch, checkable by anyone); an output that is faithfully delivered but factually wrong requires a judge, whose error rate is an irreducible cost. Colluding provider/validator pairs are mitigated structurally — deterministic, unpredictable sampling means a provider cannot know which jobs will be audited — but validator collusion at scale is **not** claimed solved. Sybil resistance is economic (identity multiplication is cheap, but acting under an identity without stake is not) rather than identity-based, and its adequacy against a well-resourced adversary is an open question. Latency misreporting is deferred to the general verification path rather than a dedicated mechanism — a stated gap.

Discovery and identity. One secp256k1 keypair doubles as network identity and settlement wallet. A slow-changing **capability record** (models served, tier, stake) propagates through a DHT; a fast-changing **liveness signal** (reachability, which models are *loaded and ready*) propagates directly between peers — kept separate because collapsing them forces a choice between an expensive high-frequency DHT update or a stale signal an auction that clears in seconds cannot use.

Market protocol. A sealed-bid, second-price *procurement* auction: providers bid the price they want to be paid; the lowest eligible bid wins but is paid the second-lowest eligible price, making truthful bidding

dominant. Eligibility (tier, latency, price ceiling) is checked as hard constraints before any price comparison. A warm-model discount is applied to *ranking score* only, never to payment — preserving individual rationality — because over realistic latency budgets, a cold provider's fixed load cost makes warmth closer to an eligibility condition than a tradeable quality dimension (confirmed empirically in §5.2).

Inference contract. A narrow, runtime-agnostic contract: stream tokens, report true time-to-first-token and a real per-request token count, and report cold/warm state. A provider that cannot report these honestly cannot participate.

Verification protocol. Every job is committed to a content-addressed store. A deterministic-but-unpredictable sample is audited: first a purely mechanical check (does the committed output match the store, under a Merkle proof requiring no model or judgment — a **fraud proof**, treated as certain); only if that passes does one or more validators render a semantic verdict, required to be one of three values — pass, fail, or **unavailable** — so a judge outage is never conflated with either an acquittal or a slash. Validator *diversity* is recorded as a first-class property rather than assumed.

Settlement. A fixed challenge window follows commitment; uncontested release requires no privileged party online. Confirmed fraud (proof or FAIL verdict) returns escrow to the requester and slashes provider collateral 80/20 between reporter and treasury (Figure 3). Withdrawn collateral stays slashable during an unbonding delay, closing a front-running window.

IV. IMPLEMENTATION

Five cooperating tracks share one wire contract, in Python except settlement: discovery/market over a Kademlia+GossipSub library (py-libp2p); inference against a streaming open-weight runtime exposing token-level timing (Ollama, replaceable behind the same contract); settlement as four Solidity contracts on a local development chain; weight distribution against a real IPFS daemon with client-side content-identifier recomputation (never trusting the daemon's own

claim). Every message carrying economic consequence is signed and independently re-verified.

TABLE I. DESIGN VERSUS PROTOTYPE

Component	Prototype	Production target	Limitation left open
Settlement chain	Local single-node dev chain	Arbitrum Stylus rollup	No fee market, no finality delay; fraud-proof

Component	Prototype	Production target	Limitation left open
			censorability untested
Data availability	Local Merkle-committed blob store	Celestia (DA sampling)	Binding is real; availability under deletion is not
Inference runtime	CPU-only, single 2B model	GPU-accelerated serving stack	No figure bears on GPU-tier economics
Verification judge	Small locally-hosted (deployed); larger hosted (measured separately)	Locally-hostable, adequately-capable pool	Deployable config fails adversary-favored strategies; the config that

Component	Prototype	Production target	Limitation left open
			succeeds isn't edge-hostable
Economic stake	Test-denomination collateral	Real, market-priced	No conclusion about deterrent adequacy

Every experiment ran on one host (16 cores, ~31 GB RAM, no accelerator); the auction was also run in per-container network namespaces with `tc netem`-injected link delay, isolating a per-peer link delay that a shared-loopback topology cannot express. Every run writes a timestamped, never-overwritten directory with a full config snapshot and git SHA; a component that cannot be reached raises a named exception rather than a plausible-looking default.

V. EVALUATION

A. Latency

Warm time-to-first-token: mean 609.6 ms, median 587.9 ms, p95 723.6 ms (n=20; **all 20 under one second**). Cold TTFT averaged 7,963.8 ms against a paired warm 653.7 ms — a **12.18×** ratio, attributable almost entirely to model *load* time (7,418 ms cold vs. 568 ms warm), not generation — the direct empirical justification for pricing warmth near eligibility rather than as a smooth quality signal.

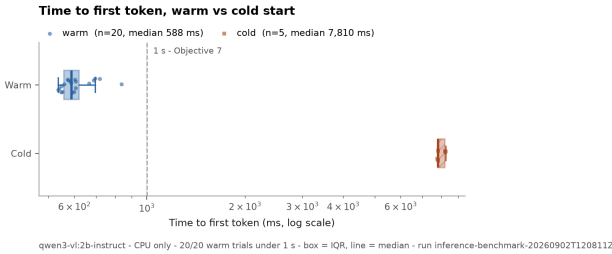


Figure 1. Time to first token, warm versus cold start, on a logarithmic time axis.

B. Auction Convergence

Fifty-seven single-machine auctions (3/4/5 nodes) cleared with zero failures; broadcast-to-award was pinned at ~2,008 ms by the fixed bid window (not a scaling signal), while un-pinned bid-arrival times carried the real signal: last-bid latency rose from 21.3 → 32.6 → 36.7 ms. Under container network namespaces with injected one-way link delay (0/10/25/50 ms), first-bid arrival rose 6.0 → 44.5 → 71.0 → 114.0 ms, close to but above a bare round-trip reference line — roughly **2.06 ms of added delay per ms injected**, consistent with GossipSub relaying through a third peer on some messages.

C. Verification Accuracy

100 trials (80 fraudulent across four corruption strategies, 20 honest) against TruthfulQA. The judge failed **none** of 20 honest answers (100% precision, 0% FPR) — a corrected result: an earlier run had shown a 75% false-positive rate, since diagnosed as a fault in the honest-answer *generator*, not the judge. **Overall recall was 65%, sharply bimodal**: 100%/95% on off-topic/fabricated-entity fraud, versus **35%/30%** on plausible-substitution/negation fraud — scored *confidently* (3.80–4.05/5 against a threshold of 3), not near-threshold, ruling out "raise the threshold" as a fix.

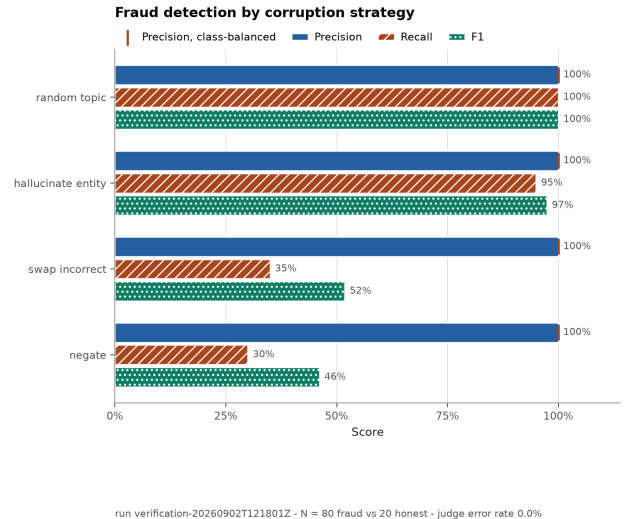


Figure 2. Fraud detection by corruption strategy: precision, recall, and F1.

Capability, not lineage. Re-judging the same 80 strings with larger hosted models: a ~13×-larger *same-family* judge recovered negation recall to 100% (98%

overall); an *unrelated* family reached 100%/90% (96% overall) — refuting our own initial hypothesis that lineage, not capability, explained the gap. Under majority-vote quorum across all four panel members, recall reached 97% at a 7% false-positive rate versus 26%/16% for individual complete members — direct support for quorum aggregation, qualified by two of four members being rate-limited into near-uselessness. Judge self-consistency under paraphrase (small judge, 8 questions): a 25% verdict-flip rate on reworded claims.

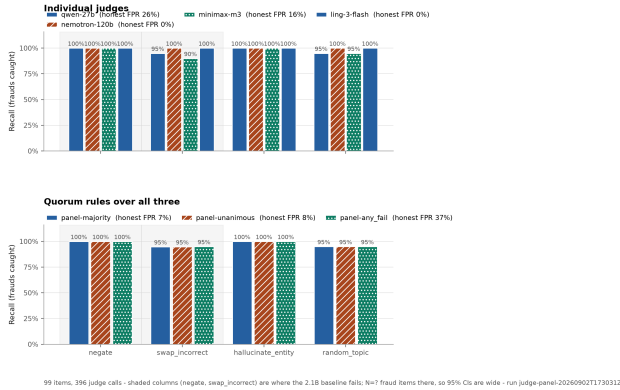


Figure 3. Recall by corruption strategy for each judge and quorum rule.

D. Weight Distribution and Cost

Five artefacts (64 KiB–48 MiB) via a real IPFS daemon: cold fetch 6.6–317.5 ms, warm (cached) under ~1.5 ms — speedups of **12.3×–895.8×** — with the content identifier independently recomputed on every fetch and three tampering scenarios correctly rejected against one honest control accepted. Three on-chain settlements (honest-release, fraud-proof-slash, verdict-slash) all resolved correctly with an exact 80/20 split and exact value conservation; gas ranged 32,317 (withdrawal) to 221,353 (the trustless fraud-proof — the best value in the system, since it requires no model). Composed cost: **\$0.00115** per 1,000 tokens (verification 4.8% of that) against a **\$0.002** centralized baseline — a notional-rate cost *model*, not a market observation.

VI. DISCUSSION

The prototype convincingly shows that the five-stage pipeline of Section III holds together as a running system rather than as a set of separately-plausible components: a real signed job travels from broadcast, through a real gossip-mesh auction that clears correctly and is individually rational on every measured award, through a real streaming inference call whose warm latency clears a sub-second target, through a real content-addressed weight fetch a node can verify without trusting its source, through a real Merkle-committed data-availability check, to a real on-chain escrow that resolves correctly down all three of its state-machine paths and conserves value exactly. That composition — not any individual mechanism, all of which are prior art (§II) — is what this evaluation is actually able to support at small scale. What it does not show is comparably important: every network measurement, including the auction under injected latency, ran on one physical kernel; nothing here establishes behavior across real wide-area links, NAT traversal, packet loss, or peer churn; three and five node counts cannot establish a scaling law; gas figures are priced against no fee market or contention;

and staked collateral in every run is a test value with no real cost to lose, so no conclusion follows about deterrent adequacy — the mechanism's incentive-compatibility argument is a property of its construction, not something measured here.

The **central finding** is not a number but a corrected explanation: it would have been easy to conclude that a same-family judge inherits the fraud-generating model's blind spots, making diversity the fix. The panel experiment refutes this — capability, not lineage, closed the gap. A second, general finding travels alongside it: **every false-positive rate in this evaluation is confounded by the fallibility of whichever model generated the "honest" control**, a milder recurrence of the exact failure that produced the earlier 75%-false-positive result. Judge self-consistency under paraphrase compounds both: a single verdict is not a stable basis for an irreversible slash. Automated, on-chain slashing driven by an LLM's judgment therefore makes *which judge, at what capability, under what quorum rule* a safety-critical configuration choice, not a tuning parameter.

VII. CONCLUSION AND FUTURE WORK

Five mechanisms studied separately in prior literature — P2P discovery, an auction market,

streaming edge inference, two-tier verification, and staked settlement — compose into one measured, end-

to-end pipeline. The prototype's most useful result is a refuted hypothesis: the verification layer's failure on adversary-favored fraud tracks judge capability, not lineage, correcting our own initial explanation rather than confirming it.

Future work, each tied to a named limitation: (1) a genuine multi-machine deployment — the largest unaddressed threat, since every network measurement here shares one kernel; (2) a capability-adequate, *edge-hostable* judge, since the configuration that closes the recall gap is not deployable on the hardware

tier this network targets; (3) a human-adjudicated honest control set, the precondition for any clean judge-precision measurement; (4) the production substitutions of Table I — Celestia, a real rollup, GPU-accelerated serving, a public IPFS swarm; (5) an adaptive adversary that reallocates toward undetected fraud strategies; (6) zero-knowledge verification of model execution, which would make judge capability moot but remains orders of magnitude too expensive for a sub-second latency target today.

References

- [1] E. Lui and J. Sun, "Bittensor protocol: The Bitcoin in decentralized AI?" *MARBLE 2025*, Springer, pp. 145–165.
- [2] G. Osuri and A. Bozanich, "AKT: Akash network token & mining economics," Whitepaper, 2020.
- [3] Golem Factory GmbH, "The Golem project: Crowdfunding whitepaper," 2016.
- [4] Gensyn AI Ltd., "Gensyn litepaper," 2022.
- [5] Morpheus, Trinity, Neo, "Morpheus: A network for powering smart agents," Whitepaper, 2023.
- [6] MorpheusAIs, *Morpheus-Lumerin-Node* [Software], 2024-.
- [7] DGrid.AI, "DGrid AI: The decentralized AI inference network," Litepaper, 2025.
- [8] C. Tong et al., "Parallax: Efficient LLM inference service over decentralized environment," arXiv:2509.26182, 2025.
- [9] H. Liu et al., "PolyLink: A blockchain based decentralized edge AI platform for LLM inference," *IEEE Blockchain 2025*, pp. 101–108.
- [10] M. Al-Bassam et al., "Fraud and data availability proofs: Detecting invalid blocks in light clients," *FC 2021*, LNCS 12675, pp. 279–298.
- [11] L. Bousfield et al., "Arbitrum Nitro: A second-generation optimistic rollup," Offchain Labs, Whitepaper, 2022.
- [12] J. Teutsch and C. Reitwießner, "A scalable verification solution for blockchains," TrueBit Whitepaper, 2017.
- [13] L. Zheng et al., "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena," *NeurIPS 2023*.
- [14] S. Lin et al., "TruthfulQA: Measuring how models mimic human falsehoods," *ACL 2022*, pp. 3214–3252.
- [15] W.-L. Chiang et al., "Chatbot Arena: An open platform for evaluating LLMs by human preference," *ICML 2024*, PMLR 235.
- [16] A. Borzunov et al., "Petals: Collaborative inference and fine-tuning of large models," *ACL 2023 Demos*, pp. 558–568.
- [17] A. Borzunov et al., "Distributed inference and fine-tuning of large language models over the Internet," *NeurIPS 2023*.
- [18] Y. Yang et al., "Navigator: A decentralized scheduler for latency-sensitive AI workflows," *IEEE EDGE 2024*, pp. 35–47.